NEWTON SCHOOL

ANSWER KEY

Multi-Step Equations with Distribution & Word Problems

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Part A: Multi-Step Equations with Distribution (1-10)

Problem 1

$2(1-a) = -(1-5a)+3$

Distributive Property | Combining Like Terms

$2-2a = -1+5a+3$
$2-2a = 2+5a$
$2-2 = 5a+2a$
$0 = 7a$

$a = 0$

Problem 2

$-2(2k+5) = -2(k-4)$

Distributive Property

$-4k-10 = -2k+8$
$-4k+2k = 8+10$
$-2k = 18$

$k = -9$

Problem 3

$5(2-4n) = -5(5n+4)$

Distributive Property

$10-20n = -25n-20$
$-20n+25n = -20-10$
$5n = -30$

$n = -6$

Problem 4

$-2(v+2)+8 = \frac{2(2+v)-4v}{2}$

Distribution | Fractions

Left: $-2v-4+8 = -2v+4$
Right: $\frac{4+2v-4v}{2} = \frac{4-2v}{2} = 2-v$
$-2v+4 = 2-v$
$-2v+v = 2-4$
$-v = -2$

$v = 2$

Problem 5

$3(2x-1) = 2(x+5)-3$

Distributive Property

$6x-3 = 2x+10-3$
$6x-3 = 2x+7$
$6x-2x = 7+3$
$4x = 10$

$x = \frac{5}{2}$ or $x = 2.5$

Problem 6

$-4(3m+2) = -2(6m+4)$

Identity Equation

$-12m-8 = -12m-8$
$-12m+12m = -8+8$
$0 = 0$ (Always true)

All real numbers (Identity)

Part A (continued)

Problem 7

$7(p-3)+2=4(p+1)-5$

Distributive Property

$7p-21+2=4p+4-5$
$7p-19=4p-1$
$7p-4p=-1+19$
$3p=18$

$p=6$

Problem 8

$-3(2y+4)=6(1-y)$

No Solution

$-6y-12=6-6y$
$-6y+6y=6+12$
$0=18$ (False)

No Solution (Contradiction)

Problem 9

$5(w-2)-3(w+1)=2(w-4)$

Identity

$5w-10-3w-3=2w-8$
$2w-13=2w-8$
$-13=-8$ (False)

No Solution (Contradiction)

Problem 10

$\frac{3(x+2)}{3}=\frac{2(x-1)+8}{2}$

Simplify Fractions

$x+2=\frac{2x-2+8}{2}$
$x+2=\frac{2x+6}{2}=x+3$
$2=3$ (False)

No Solution (Contradiction)

Part B: Word Problems (11-20)

Problem 11

> The perimeter of a rectangle is 332 cm and the width is 76 cm. Find the rectangle's length.

Perimeter | Rectangle

Let l = length

Perimeter formula: $2l + 2w = P$

$2l + 2(76) = 332$

$2l + 152 = 332$

$2l = 180$

Length = 90 cm

Problem 12

> The perimeter of a rectangle is 408 cm and the length is 134 cm. Find the rectangle's width.

Perimeter | Rectangle

Let w = width

$2(134) + 2w = 408$

$268 + 2w = 408$

$2w = 140$

Width = 70 cm

Problem 13

> Isosceles triangle has perimeter 345 cm and base length 85 cm. Find each leg.

Isosceles Triangle | Perimeter

Let x = length of each leg

$2x + 85 = 345$

$2x = 260$

Each leg = 130 cm

Part B (continued)

Problem 14

> Theo has $5 more than Denise and Denise has $11 more than Rudy. Together they have $45.

System Setup | Money Problem

Let r = Rudy's amount

Denise = $r + 11$, Theo = $r + 11 + 5 = r + 16$

$r + (r + 11) + (r + 16) = 45$

$3r + 27 = 45$

$3r = 18$, so $r = 6$

Rudy = $6, Denise = $17, Theo = $22

Problem 15

> Chandra has twice as much money as Nora. Nora has $6 less than Lian. Together they have $54.

System Setup | Money Problem

Let L = Lian's amount

Nora = $L - 6$, Chandra = $2(L - 6) = 2L - 12$

$L + (L - 6) + (2L - 12) = 54$

$4L - 18 = 54$

$4L = 72$, so $L = 18$

Lian = $18, Nora = $12, Chandra = $24

Problem 16

> Machine A caps twice as many bottles as Machine B. Machine C caps 500 more than A. Total: 40,000 bottles.

System Setup | Work Problem

Let B = Machine B's bottles

A = $2B$, C = $2B + 500$

$B + 2B + (2B + 500) = 40000$

$5B + 500 = 40000$

$5B = 39500$, so $B = 7900$

B = 7,900, A = 15,800, C = 16,300 bottles

Part B (continued)

Problem 17

> A farmer has chickens and cows. Total heads: 50. Total legs: 140. Find each.

System of Equations | Classic Problem

Let c = chickens, w = cows

$c + w = 50$ (heads)

$2c + 4w = 140$ (legs)

From first: $c = 50 - w$

$2(50 - w) + 4w = 140$

$100 - 2w + 4w = 140$

$2w = 40$, so $w = 20$

Chickens = 30, Cows = 20

Problem 18

> The sum of three consecutive integers is 72. Find the three integers.

Consecutive Integers

Let integers be n, $n + 1$, $n + 2$

$n + (n + 1) + (n + 2) = 72$

$3n + 3 = 72$

$3n = 69$, so $n = 23$

23, 24, 25

Problem 19

> Length is 4 cm more than twice the width. Perimeter is 50 cm. Find dimensions.

Perimeter | Rectangle

Let w = width, length = $2w + 4$

$2(2w + 4) + 2w = 50$

$4w + 8 + 2w = 50$

$6w = 42$, so $w = 7$

Length = $2(7) + 4 = 18$

Width = 7 cm, Length = 18 cm

Problem 20

> Maria is 3 years older than twice her brother's age. Sum of ages is 18 years.

Age Problem

Let b = brother's age

Maria = $2b + 3$

$b + (2b + 3) = 18$

$3b + 3 = 18$

$3b = 15$, so $b = 5$

Maria = $2(5) + 3 = 13$

Brother = 5 years, Maria = 13 years

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